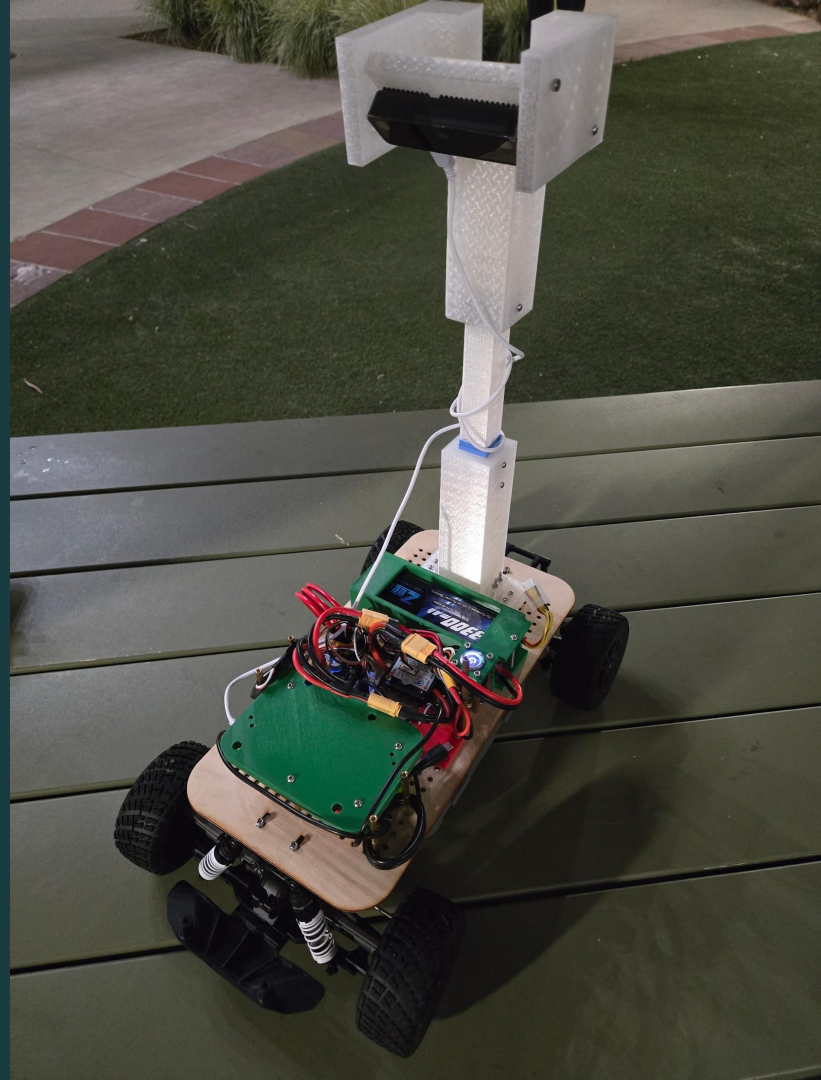


Parking Space Finder

ECE MAE 148 Team 8

Geonmoo Lee – MAE
Christopher Wong – MAE
Henry Tiet – ECE



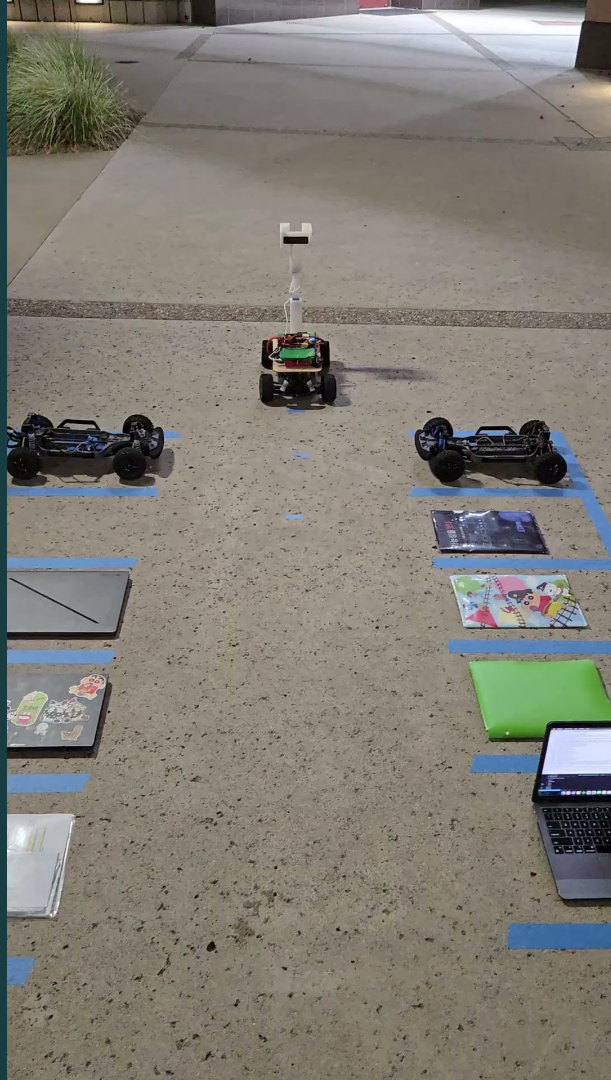
Reminder of What We Promised

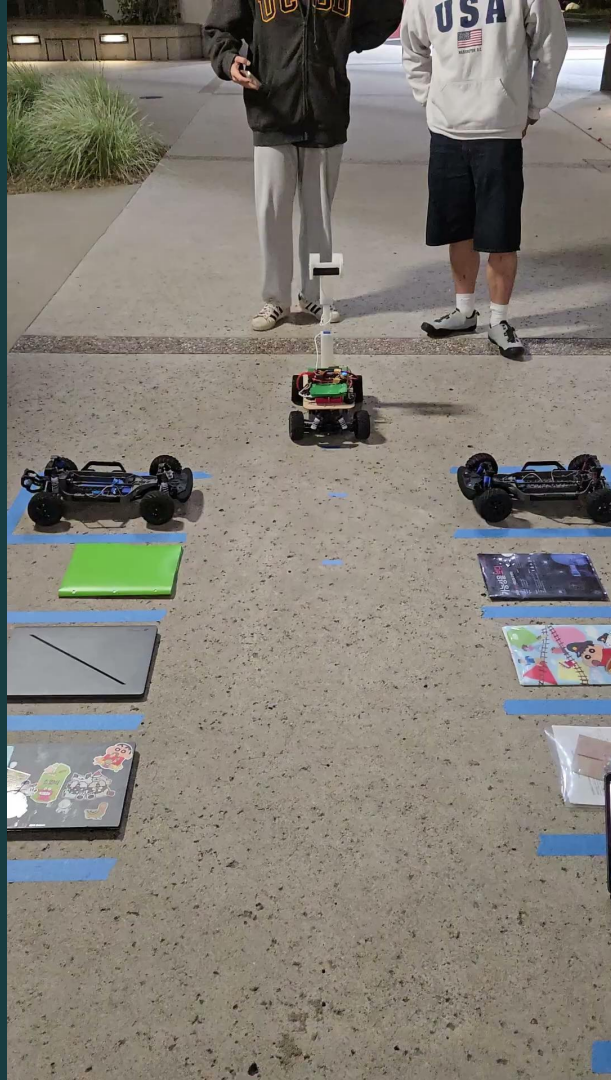
Must Haves:

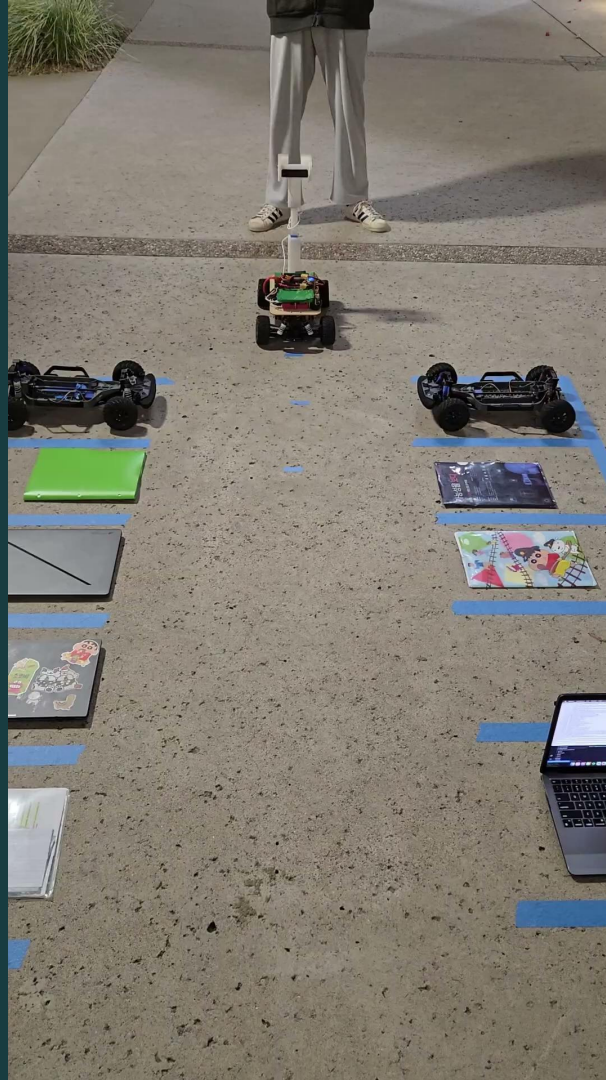
- Camera Recognition of an Open Parking Spot
- Vesc Control of Motor to Park into Spot
- Avoidance of Blue Tape Lines that Mark the Parking Lot

Nice to Have:

- SLAM Mapping of Parking Lot
- Identification of Different-sized Obstacles
- Understanding Badly-parked Neighbors and Avoidance of Them







What We Have Done

What Worked:

- API call to Roboflow for YOLO model
- Camera Output Changes (Masking, Exposure, Hue, ISO)
- Communication between VESC and PI for auto parking

What didn't Work as Expected and Why:

- 1) Training an Onboard YOLO model with high enough confidence – Not enough photos in the database (200) OR Roboflow has unique weights to its training model that we didn't have
- 2) Odometry – precise control was still difficult due to misalignment of the wheels

How can we Solve the Issue(s):

- 1) Adding more photos with good annotations should have allowed for a better model and higher confidence (annotating the taken spaces may also have helped)
- 2) Extra time working with ROS2 and a bit more trial and error should have enabled us to properly engage with the program, we used a predetermined set of timed input pulses as a bandaid solution

If we had another week

Extend our Existing Parking Lot

- Add a loop and curve the parking lot for a bigger section

Add path following for more Complicated Parking Lots

- With a curvy parking lot, path following will be required to not cross over any parking lines

Train a Working YOLO model for Onboard Processing

- Our currently model API calls back to Roboflow due to multiple YOLO models having too low of confidence in identifying parking spots (less than 0.05)

Make Parking More Robust

- Use the camera while parking to ensure that the parking is accurate and doesn't get off track, like ensuring the car is parked fully into the parking spot (from using the blue line in front of it) and that the car doesn't go over the parking lines

UC San Diego

JACOBS SCHOOL OF ENGINEERING

Gantt Chart Update

Tasks	Week 7	Week 8	Week 9	Week 10	Finals Week
Hardware Test					
Redesign the Camera Mount					
Build Parking Lot					
Train YOLO/AI Model					
Testing For Occupancy					
Creep Integration					
Final Parking					

UC San Diego

JACOBS SCHOOL OF ENGINEERING

Lessons Learned

- 1) For any image-based models, try and have consistent lighting or have multiple photos of differing lightings
- 2) The amount of images needed for a YOLO model is much higher than you think
- 3) For debugging, program an update output for the program to display for instant feedback
- 4) Allot more time for each and every step as you will need at the least 1.5x the amount of time you think you need for project steps
- 5) Charge the battery at the end of using it so that next time there isn't a need to start with charging the battery
- 6) Don't leave the battery plugged in overnight (even the battery monitor), even when the robot is not powered on (We learned the hard way)
- 7) remember: This class is suppose to be fun. You are working hard to add skilss to your resume not just to pass the class.

Go!

Jack (edited)